Practice Problems: Maxima, Minima and Monotonicity

Course: Pre-Calculus-I

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Instructions

- Solve **without** using derivatives.
- Use tables, graphs, completing the square, range bounds, or the peak/valley test.
- On closed intervals, always compare interior extrema with **endpoint values**.
- State monotonicity as intervals, e.g. increasing on (a, b).

Note 1. *Problems are arranged in increasing order of difficulty. P1–P4 are warm-up; P5–P8 are core; P9–P12 are challenge.*

1 Easy (P1–P4)

P1. For $f(x) = -4x + 9$ on $[-1, 3]$:

(a) State the monotonicity on $\mathbb{R}$.

(b) Find the absolute maximum and absolute minimum on $[-1, 3]$.

P2. Let $f(x) = x^2 - 6x + 5$.

(a) Complete the square and state the vertex.

(b) Find the absolute extremum on $\mathbb{R}$ and its type.

(c) State the intervals of increase and decrease.

P3. For $f(x) = 3 - 2|x + 1|$:

(a) Find the absolute maximum on $\mathbb{R}$ and where it occurs.

(b) State the intervals of monotonicity.

(c) Find the absolute extrema on $[-3, 2]$.

P4. Show that $f(x) = \left(\frac{1}{3}\right)^x$ is decreasing on $\mathbb{R}$ by comparing $f(x_1)$ and $f(x_2)$ for $x_1 < x_2$. State its range.

2 Medium difficulty (P5–P8)

P5. Using a table of values on $x \in \{-2, -1, 0, 1, 2, 3\}$, determine the intervals of increase/decrease and all local extrema of

$$f(x) = x^3 - 3x^2.$$

Confirm the peaks/valleys using the peak/valley test.

P6. Find the absolute extrema of $f(x) = -x^2 + 4x + 1$ on $[0, 5]$.

(a) Complete the square first.

(b) List all candidate points (vertex and endpoints) with values.

(c) State the absolute maximum and minimum with their locations.

P7. For $f(x) = 2\sin\left(x - \dfrac{\pi}{4}\right) + 3$ on $[0, 2\pi]$:

(a) State the amplitude, vertical shift, and range.

(b) Find the absolute maximum and minimum values on $[0, 2\pi]$ and the x-values where they occur.

P8. The profit (in thousands) of a company is modelled by

$$P(x) = -3x^2 + 36x - 60, \qquad x \geq 0,$$

where x is the number of units (in hundreds) produced.

(a) Complete the square to find the vertex.

(b) Determine the production level that maximises profit and the maximum profit.

(c) For which $x \geq 0$ is the profit increasing?

3 Challenging Problems (P9–P12)

P9. Let $f(x) = x^4 - 4x^2$.

(a) Tabulate f at $x = -2, -1, 0, 1, 2$ and at $x = \pm\sqrt{2}$.

(b) Identify all peaks and valleys of the graph.

(c) State the intervals of increase and decrease on $\mathbb{R}$.

(d) Find the absolute minimum of f on $\mathbb{R}$; does f have an absolute maximum on $\mathbb{R}$? Justify.

P10. Consider $g(x) = e^{-x^2}$.

(a) Use the bound $x^2 \geq 0$ to show $0 < g(x) \leq 1$ for all x.

(b) Where does g attain its absolute maximum? Justify without using derivatives.

(c) Show that g is increasing on $(-\infty, 0)$ and decreasing on $(0, \infty)$ by comparing $g(x_1)$ and $g(x_2)$ for $x_1 < x_2$ in each interval.

P11. Let $f(x) = |x - 2| + |x + 2|$.

(a) Write f as a piecewise function with breakpoints at $x = \pm 2$.

(b) Simplify f on each of the three intervals.

(c) Find the absolute minimum of f on $\mathbb{R}$ and all x where it is attained.

(d) State the intervals of monotonicity of f on $\mathbb{R}$.

P12. Let $h(x) = \dfrac{x^2 - 4x + 5}{x - 2}$ for $x \neq 2$.

(a) Rewrite $h(x)$ in the form $(x - 2) + \dfrac{1}{x - 2}$.

(b) Using the bound $\left(t + \dfrac{1}{t}\right) \geq 2$ for $t > 0$ and $\left(t + \dfrac{1}{t}\right) \leq -2$ for $t < 0$, find the **local** maximum and **local** minimum of h.

(c) State the intervals on which h is increasing and decreasing.

(d) Does h have absolute extrema on its domain? Justify.

Selected Hints

- **P3:** $-2|x + 1| \leq 0$, so $f(x) \leq 3$ with equality at $x = -1$.
- **P5:** Factor $f(x) = x^2(x - 3)$; peaks/valleys occur between sign changes of consecutive differences.
- **P7:** The sine factor reaches ± 1 at specific x-values; check which lie in $[0, 2\pi]$.
- **P9:** Substitute $u = x^2$ to see $f = u^2 - 4u = (u - 2)^2 - 4$.
- **P10:** $e^{-x^2} = 1/e^{x^2}$; the exponent $-x^2$ is maximised at $x = 0$.
- **P11:** On $[-2, 2]$, $f(x) = 4$; outside, $f(x) = 2|x|$.
- **P12:** Let $t = x - 2$; then $h = t + 1/t$. The bound gives extrema at $t = \pm 1$, i.e. $x = 3$ and $x = 1$.

Note 2. *Attempt every problem on your own before checking the hints.*